

What are the hot sectors for new clients

According to the people interviewed for this article, mobile manufacturing has picked up big time in India. Mobile-battery testing is not something new but there were very few players before. Now almost everyone who is doing mobile manufacturing has to, and wants to, test batteries. The industry is also seeing a growing demand from education labs, research labs and companies working on Internet of Things applications and wearables, where power efficiency is a concern.

ed SMU comes with a capacitive touch-screen graphical user interface (GUI). The press release states that it offers higher power sourcing (up to 105V, 7A DC/7A pulse, 100W maximum) with 0.012 per cent basic measurement accuracy and 6½-digit resolution, making it ideal for high-power, high-precision I-V characterisation of modern materials and high-power devices.

Targeting the LED lighting sector, Yokogawa Corp. of America is introducing a new multichannel SMU in addition to GS820 models, which feature isolated 2-channel source and measurement functions. These offer four-quadrant operation consisting of the current source and current sink. On the 50V model, voltage ranges will be 200mV, 2V, 20V and 50V. Current ranges are selectable from 200 nanoamps to one amp.

Photovoltaic power system simulation/emulation

Firms that work on photovoltaic cells, panels and solar arrays need to improve the micro-inverters in their solar panels. In order to test a good micro-inverter or solar inverter, you need to simulate cell behaviour, which is a really hard task. You might need to simulate sunny, cloudy, rainy or dark days and other similar weather changes.

An engineer using a normal power supply would find it difficult to transition from sunny to rainy. Why? You need a power supply with substantial power and high wattages, at the same time being able to modulate it from low to mid to high in a seamless manner. There are power supplies that are capable of doing this with effectively less noise, and Keysight's N89xxAPV photovoltaic array simulator is one recent launch.

The simulator drops down voltage and power levels based on sunny days going to cloudy and then to rainy without the engineer having to code.

"Previously you had to cascade power supplies. Doing that for a few supplies is fine but cascading many systems creates a lot of noise and makes it difficult to manage a bunch of boxes. Engineers need to ensure their solar inverters are capable of converting maximum power from the solar array to which these are connected. Developing and verifying the performance of complex inverter maximum power point tracking (MPPT) algorithms and circuits is also challenging," says Mukul Pareek, marketing program manager - wireless and digital at Keysight Technologies.

Tektronix has recently released Keithley 2281S battery simulator and precision DC power supply. This battery simulator is claimed to offer the industry's first emulation of battery performance from full charge to total discharge using a battery model that includes state of charge, amp-hour capacity, equivalent series resistance and open circuit voltage. "Battery simulation or power supplies with fast transient responses are designed to test battery-operated products. These can capture load-current pulses as narrow as 50µs and sleep-mode currents as low as a few µA to simulate a discharged battery and test charger control circuitry," says Narasimhan.

Incremental specification improvements

"High-density equipment is one area where we see change. The number of channels that can be packed is significantly higher than what was previously covered, and it has now reached 70 channels in a 3U 48.3cm (19-inch) rack space. In our PXI product line, we now have a 200V, which is an increase from 60V SMU," explains Chivukula.

SL Power Electronics announced its 425W power supply under the series TU425 family, which is claimed to be 90 per cent efficient and comes with

active current share for redundant applications. It is compliant to IEC61000-4 standard and claims to help reduce the time spent on isolating the measurement part of a design from external electromagnetic influences.

HAMEG instruments, a subsidiary of Rohde & Schwarz, has come out with R&S HMC804x power supplies, which can feature up to three channels and features tracking multiple channels and enables something called FuseLink. This combines safety switch-off current limits for multiple channels. "The function is especially useful if two circuits with different voltages connected to a power supply require overload protection, while at the same time a third circuit needs to continue operating, for example, for a fan to provide interruption-free cooling," states their press release.

N8900 is a new auto-ranging DC power supply that can handle 15kW per piece and in a cascade can provide up to 100kW. Maximum power output was 5kW in 2013. The increase from 5kW to 15kW was achieved without an increase in box size. This instrument allows you to define wattage or power, as well as maximum and minimum voltage and current.

Then, the power supply auto-ranges from minimum and maximum at full power. Auto-ranging saves money because it provides a wide range of power output and dynamic range from only one device. N7900 series from Keysight features an in-built dissipator unit which is in sync with the power supply. This saves the user a lot of bench space, the hazard of connections, reduction in programming requirements (you do not have to separately programme the dissipator), comes with better specs and is claimed to be the fastest bench power supply by Keysight.

NI launched PXIe-4139 that helps engineers get extended range-pulsing capability up to 500W and sensitivity down to 100fA. Additionally, NI PXIe-4139 system SMU can take measurements at 1.8MS/s, which is 100 times faster than traditional SMUs.

Teledyne Lecroy has introduced